I Sustainable Water Recycling System with Interconnected Tanks

I.1 Problem Statement

Design and control a sustainable water recycling system for an industrial plant using three interconnected tanks. The system must:

- Minimize freshwater consumption by maximizing treated water reuse.
- Maintain safe water levels in all tanks to prevent overflow/underflow.
- Balance energy efficiency (e.g., pump usage) with recycling performance.

I.2 Key Features

- **Nonlinear Dynamics:** Flow between tanks depends on the square root of water level differences (e.g., Flow $\propto \sqrt{\Delta h}$).
- **Coupling:** Tank levels are interdependent (e.g., changes in Tank 1 affect Tanks 2 and 3).
- **Actuators:**
  - Freshwater inflow valve (Tank 1) controlled by PID.
  - Recycling pump (Tank 3) controlled by relay or PID.

I.3 Modeling

I.3.a Nonlinear Differential Equations

Derive nonlinear differential equations for all tanks using mass balance principles. Example relationships:

- Flow between tanks: $Q_{i\to j} = k \cdot \text{sign}(h_i - h_j) \cdot \sqrt{|h_i - h_j|}$.
- Include freshwater inflow (Q_{in}) and recycled flow (Q_{recycle}).

For each tank, we can write the following equations:

- **Tank 1:**

$$\frac{dh_1}{dt} = \frac{1}{A_1}\left(Q_{\text{in}} + Q_{\text{recycle}} - k_1 \cdot \text{sign}(h_1 - h_2) \cdot \sqrt{|h_1 - h_2|}\right)$$

- **Tank 2:**

$$\frac{dh_2}{dt} = \frac{1}{A_2}\left(k_1 \cdot \text{sign}(h_1 - h_2) \cdot \sqrt{|h_1 - h_2|} - k_2 \cdot \text{sign}(h_2 - h_3) \cdot \sqrt{|h_2 - h_3|}\right)$$

- **Tank 3:**

$$\frac{dh_3}{dt} = \frac{1}{A_3}\left(k_2 \cdot \text{sign}(h_2 - h_3) \cdot \sqrt{|h_2 - h_3|} - Q_{\text{out}} - Q_{\text{recycle}}\right)$$

Where A_1, A_2, A_3 are the cross-sectional areas of Tanks 1, 2, and 3 respectively.

I.3.b System Parameters

Define the system parameters, such as the flow coefficients k_1 and k_2, the cross-sectional areas of the tanks A_1, A_2, A_3, and the flow rates Q_{in} and Q_{recycle}.

I.4 Discretization

I.4.a Euler Method

Discretize the system using the Euler method. The discretized equations are:

- **Tank 1:**

$$h_1(t + \Delta t) = h_1(t) + \frac{\Delta t}{A_1}\left(Q_{\text{in}} + Q_{\text{recycle}} - k_1 \cdot \text{sign}(h_1(t) - h_2(t)) \cdot \sqrt{|h_1(t) - h_2(t)|}\right)$$

- **Tank 2:**

$$h_2(t+\Delta t) = h_2(t)+\frac{\Delta t}{A_2}\left(k_1 \cdot \text{sign}(h_1(t) - h_2(t)) \cdot \sqrt{|h_1(t) - h_2(t)|} - k_2 \cdot \text{sign}(h_2(t) - h_3(t)) \cdot \sqrt{|h_2(t}\right.$$

- **Tank 3:**

$$h_3(t+\Delta t) = h_3(t)+\frac{\Delta t}{A_3}\left(k_2 \cdot \text{sign}(h_2(t) - h_3(t)) \cdot \sqrt{|h_2(t) - h_3(t)|} - Q_{\text{out}} - Q_{\text{recycle}}\right)$$

I.4.b Validation

Validate the results by comparing the continuous-time model (solved with ode45) and the discretized model (implemented in a loop).

I.5 Control Design

I.5.a PID for Freshwater Inflow

Design a PID controller to regulate the level of Tank 1. The PID gains will be tuned to achieve a stable and fast response.

- **PID Controller:**

$$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t)\,dt + K_d \cdot \frac{de(t)}{dt}$$

where $e(t) = h_1^{\text{desired}} - h_1(t)$.

I.5.b Relay/PID for Recycling Pump

Compare a relay (on/off) controller with a PID controller for Tank 3.

- **Relay Controller:**

$$u(t) = \begin{cases} Q_{\text{recycle}}^{\max} & \text{if } h_3(t) < h_3^{\min} \\ 0 & \text{if } h_3(t) > h_3^{\max} \end{cases}$$

- **PID Controller:**

$$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t)\,dt + K_d \cdot \frac{de(t)}{dt}$$

where $e(t) = h_3^{\text{desired}} - h_3(t)$.

I.5.c Testing Controllers

Test the controllers on both continuous and discretized models to evaluate their performance.

I.6 System Architecture

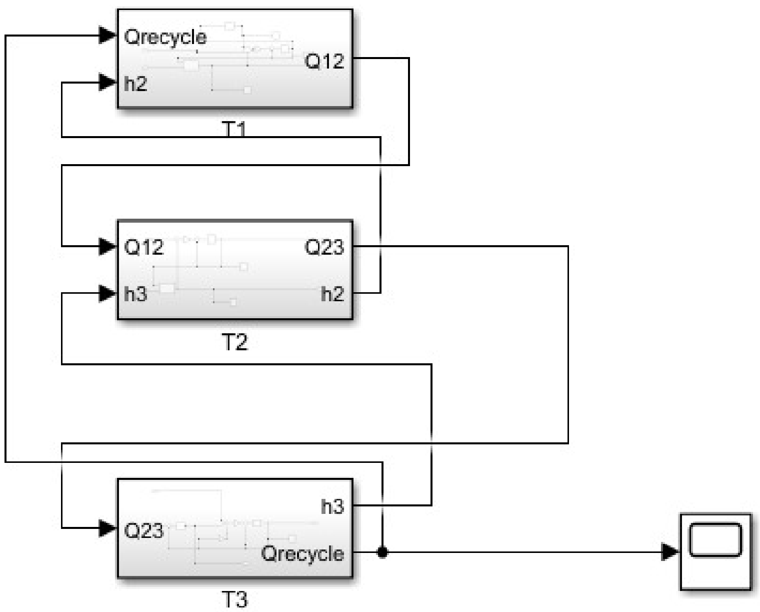

Figure I.1: General system architecture

The Simulink model consists of three masked subsystems representing Tanks 1–3 connected in series. Freshwater enters Tank 1, flows into Tank 2, then into Tank 3, and a recycling pump returns treated water to Tank 1. Each subsystem encapsulates the nonlinear mass-balance equations and allows swapping of control strategies.

I.6.a Continuous-Time PID Implementation

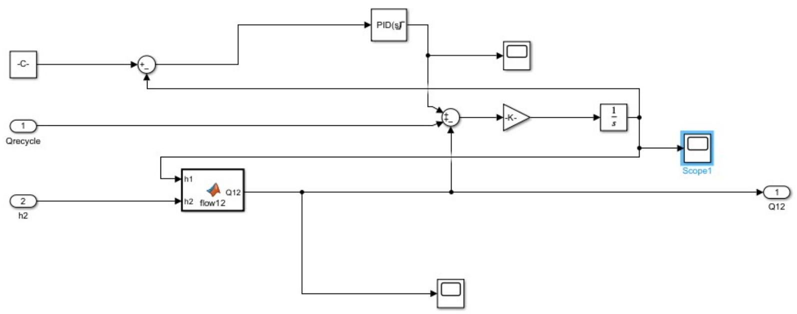

Figure I.2: Tank 1 PID block

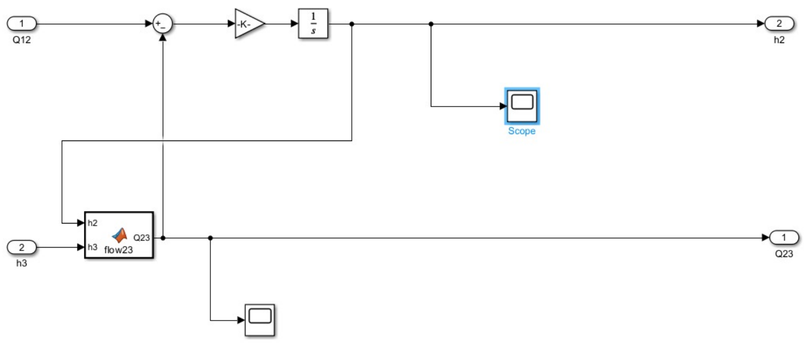

Figure I.3: Tank 2 flow dynamics

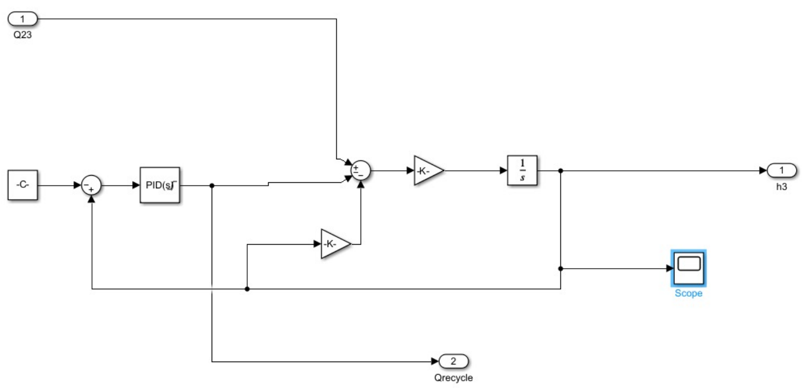

Figure I.4: Tank 3 PID block

Each PID block calculates the error between the target and measured level. In Tank 1, the PID output adjusts the inlet valve to control inflow. Tank 2 computes the nonlinear inter-tank flow using the square-root relationship but has no direct actuator. In Tank 3, the PID output drives the recycling pump to maintain the desired level.

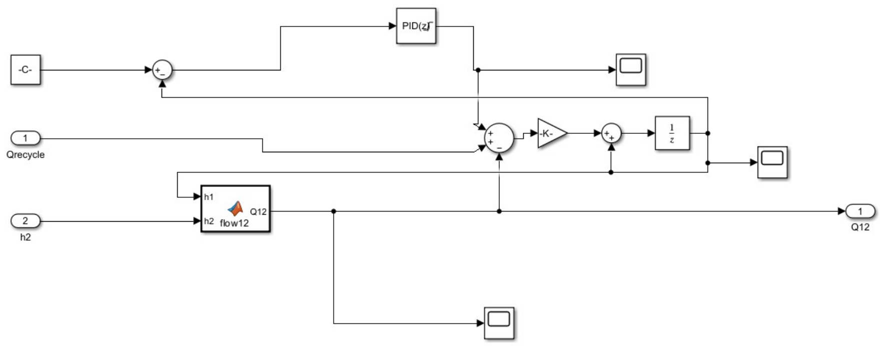

Figure I.5: Tank 1 discrete-time PID block

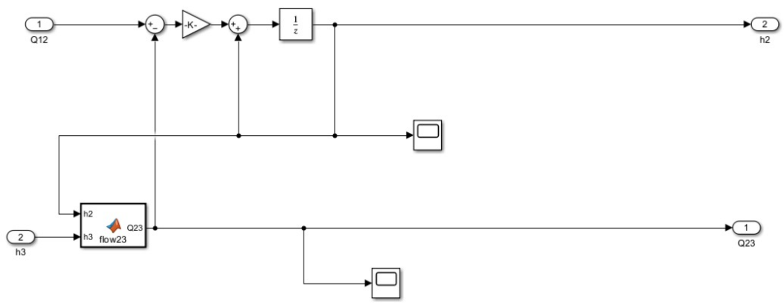

Figure I.6: Tank 2 discrete-time flow dynamics

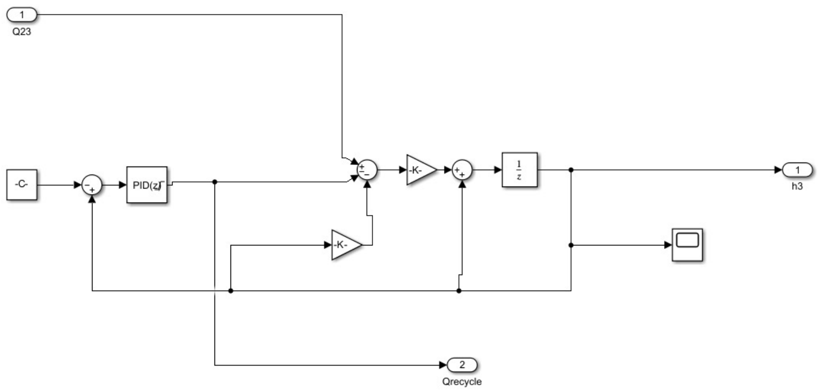

Figure I.7: Tank 3 discrete-time PID block

In this discrete-time setup, each tank uses a unit-delay element in place of the continuous integrator. Tanks 1 and 3 employ sampled-data PID controllers that act on the level error to adjust the inlet valve and the recycling pump, respectively. Tank 2 computes its nonlinear inter-tank flow based on the square-root law but has no direct actuator.

I.6.b Discrete-Time PID Implementation

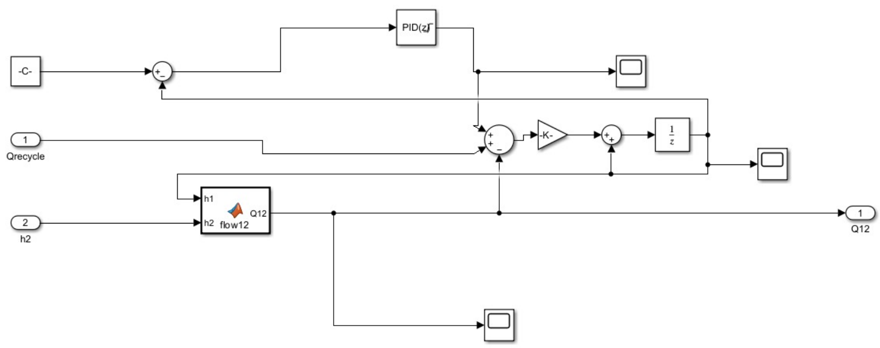

Figure I.8: Tank 1 discrete-time PID block

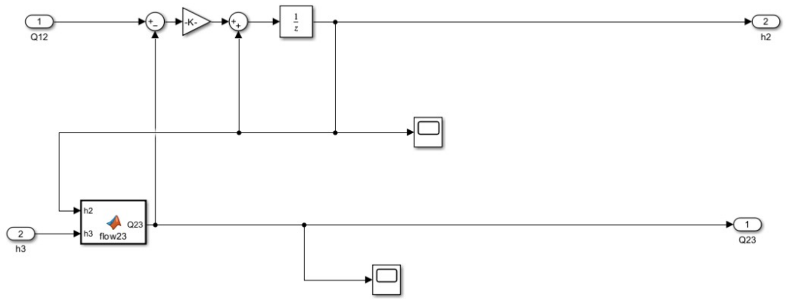

Figure I.9: Tank 2 discrete-time flow dynamics

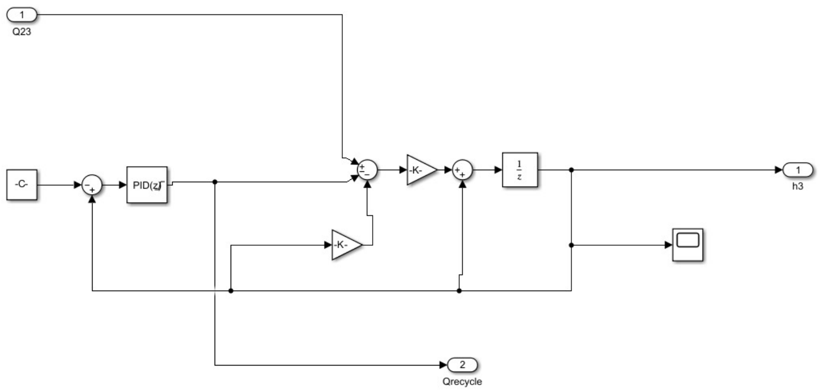

Figure I.10: Tank 3 discrete-time PID block

In this discrete-time setup, each tank uses a unit-delay element in place of the continuous integrator. Tanks 1 and 3 employ sampled-data PID controllers that act on the level error to adjust the inlet valve and the recycling pump, respectively. Tank 2 computes its nonlinear inter-tank flow based on the square-root law but has no direct actuator.

I.6.c Relay Implementation

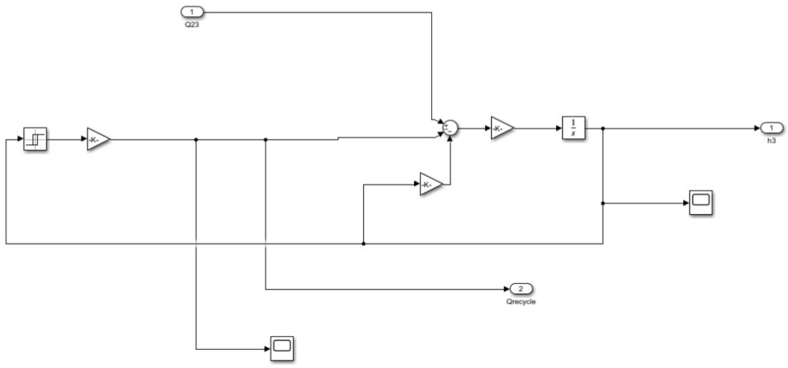

Figure I.11: Tank 3 relay block (continuous-time)

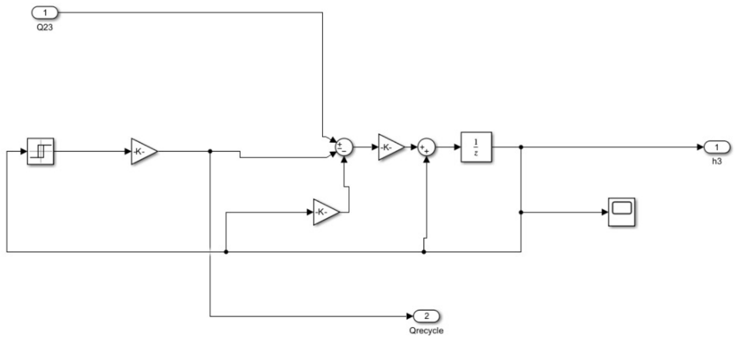

Figure I.12: Tank 3 relay block (discrete-time)

In both continuous- and discrete-time relay implementations, the block monitors the level error in Tank 3 and switches the recycling pump fully on or off whenever the level crosses its configured thresholds. The continuous version uses a comparator directly feeding the pump gain, while the discrete version inserts a one-step delay before the comparator to match the sampling interval.

I.7 Simulation Results

I.7.a Continuous-Time PID Step Responses

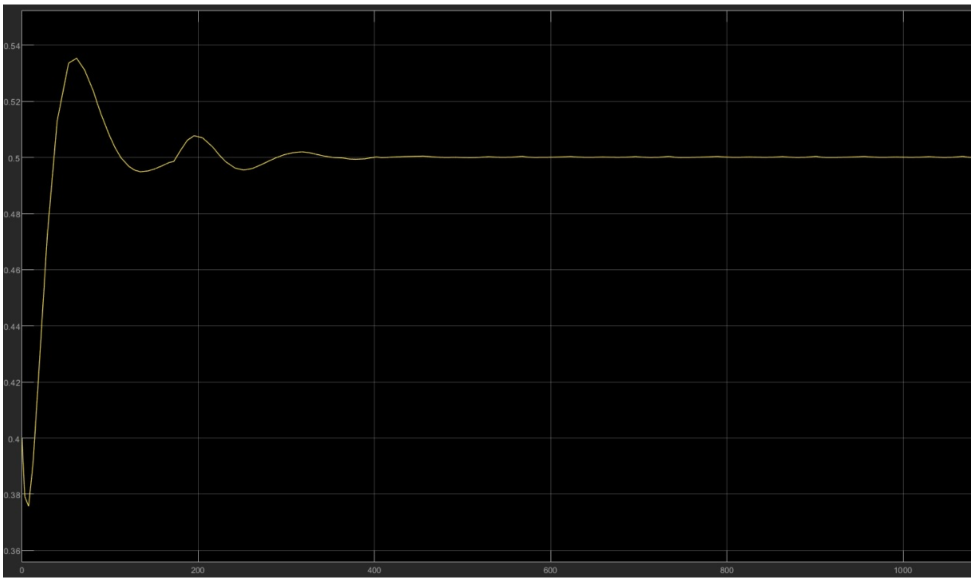

Figure I.13: Tank 1 level response

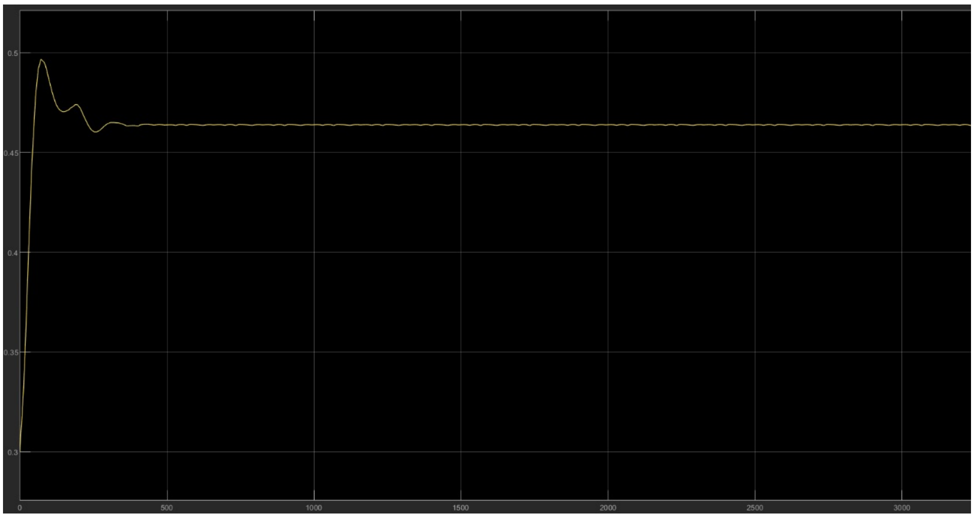

Figure I.14: Tank 2 level response

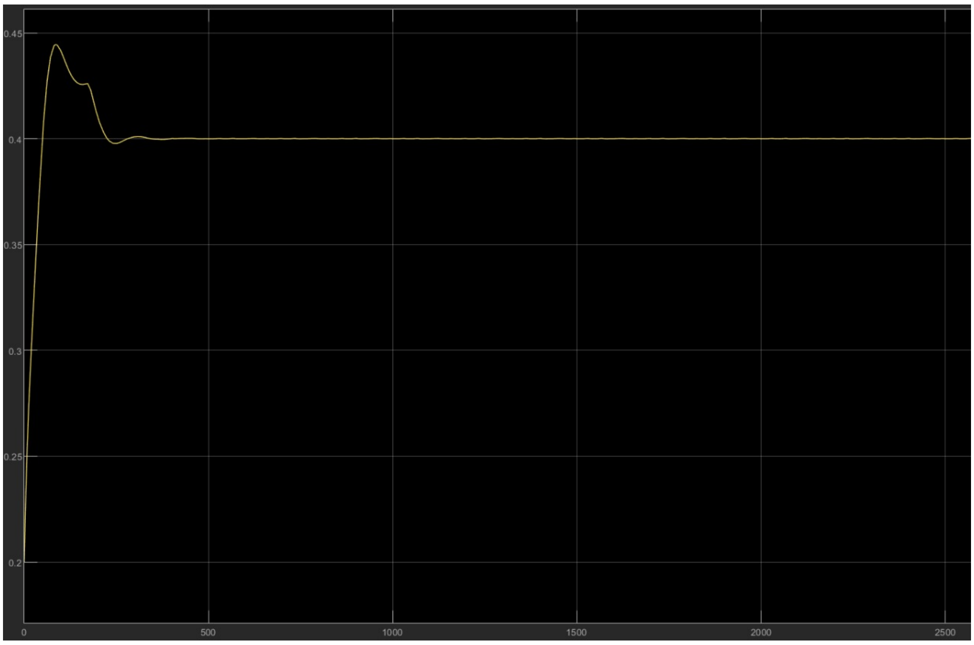

Figure I.15: Tank 3 level response

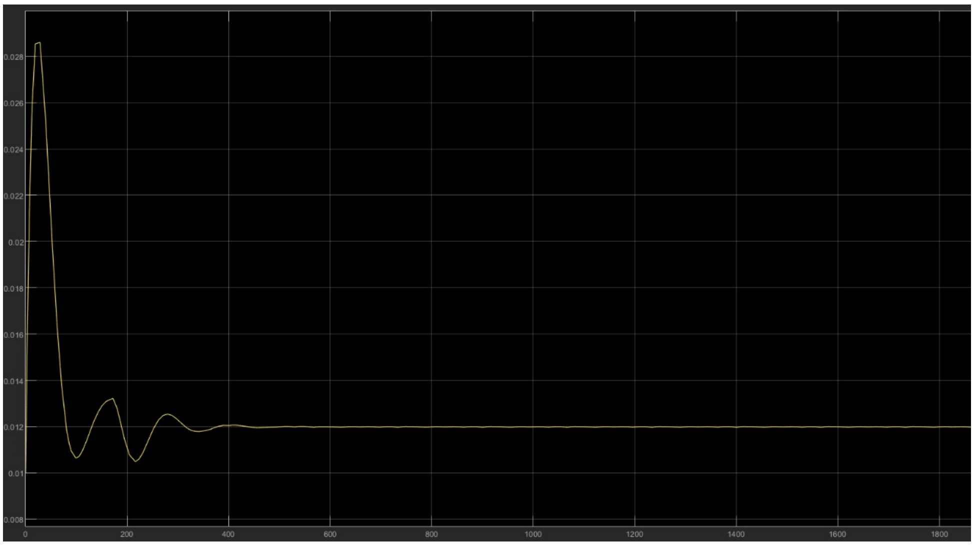

Figure I.16: Freshwater inflow

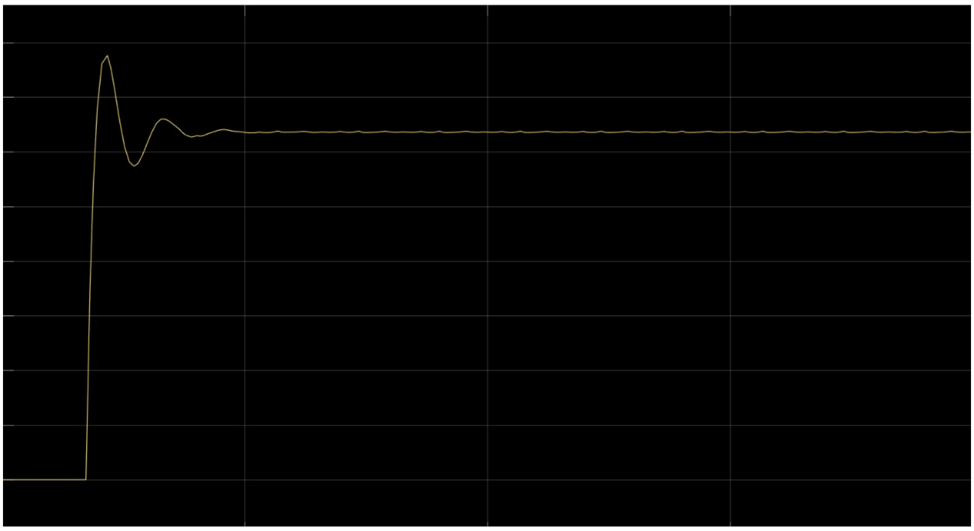

Figure I.17: Recycling flow

The step-response figures illustrate the dynamic behavior after a sudden change in level setpoint at $t = 0$. Figure I.13 shows that Tank 1's level rises quickly, overshoots to about 0.54 m, and then settles at 0.50 m. Figure I.14 reveals a similar transient in Tank 2, with an overshoot near 0.50 m before converging to 0.47 m. Figure I.15 indicates that Tank 3 peaks around 0.45 m and then smoothly reaches 0.40 m. The corresponding control actions are shown in Figures I.16 and I.17: the freshwater inflow Q_{in} spikes initially to correct the error and then remains constant, while the recycling flow Q_{recycle} pulses to about 0.028 m^3/s before declining and stabilizing near 0.012 m^3/s.

I.7.b Discrete-Time PID Step Responses

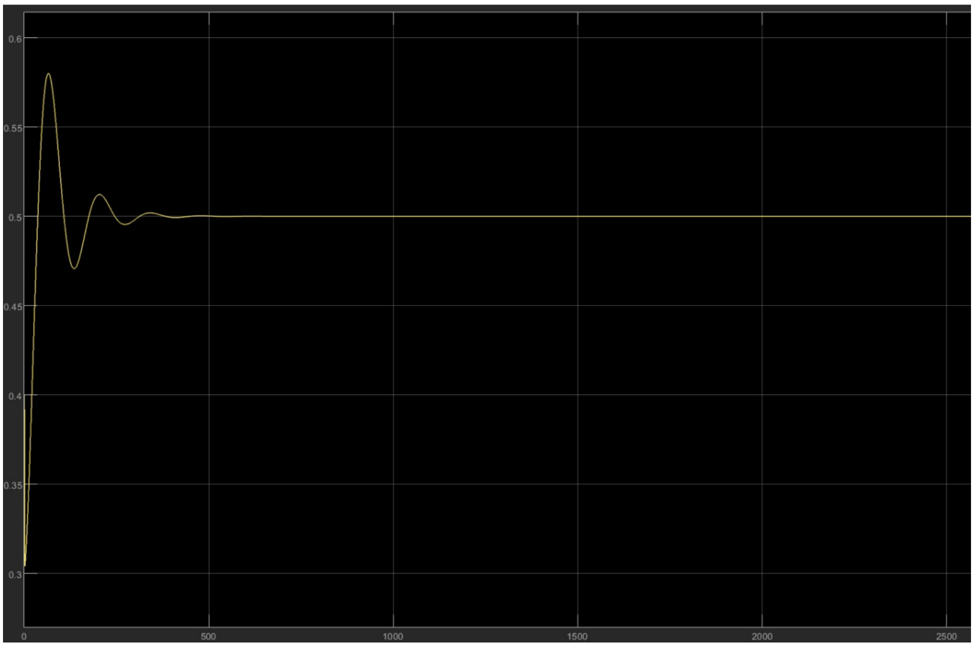

Figure I.18: Tank 1 level response (discrete-time)

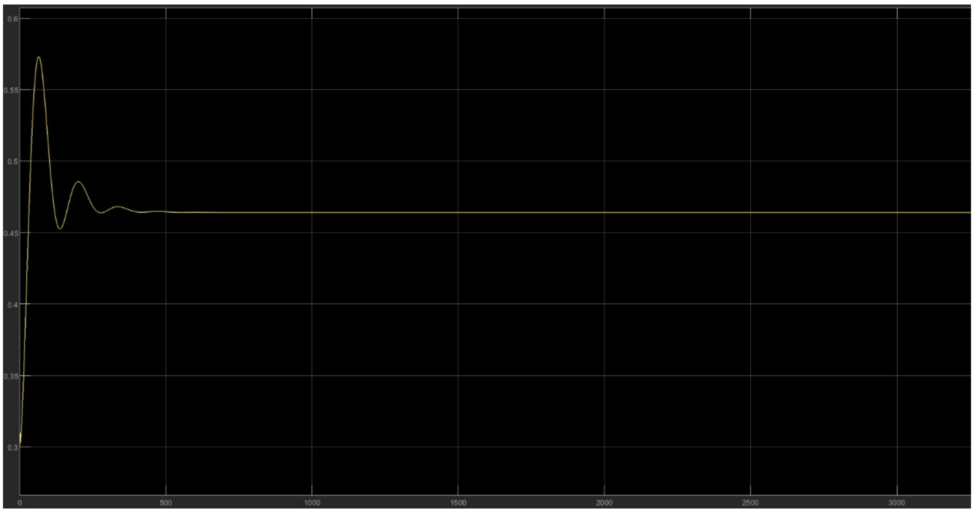

Figure I.19: Tank 2 level response (discrete-time)

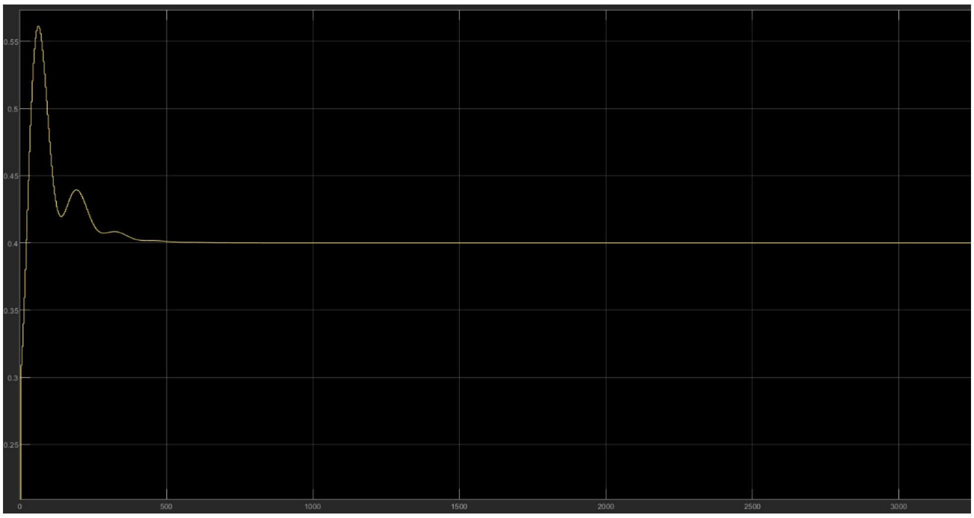

Figure I.20: Tank 3 level response (discrete-time)

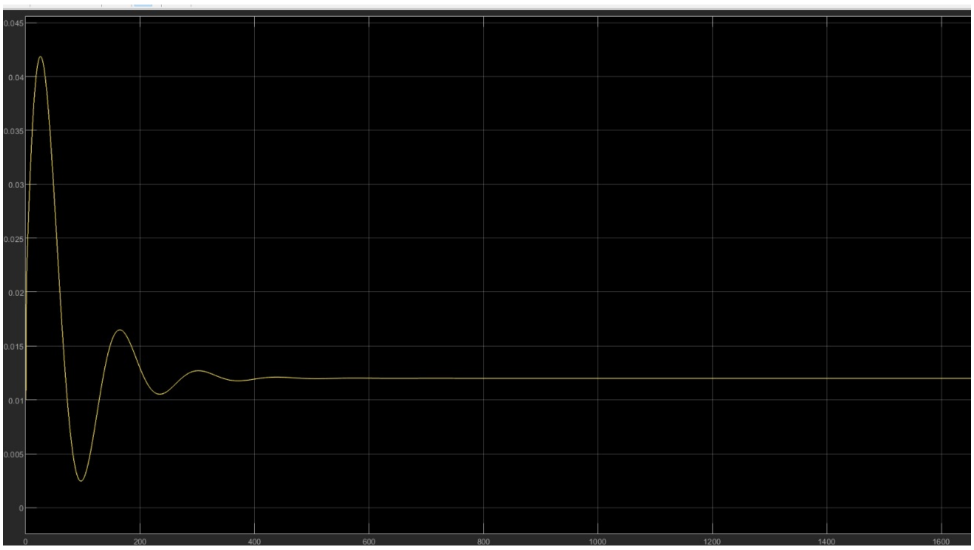

Figure I.21: Freshwater inflow (discrete-time)

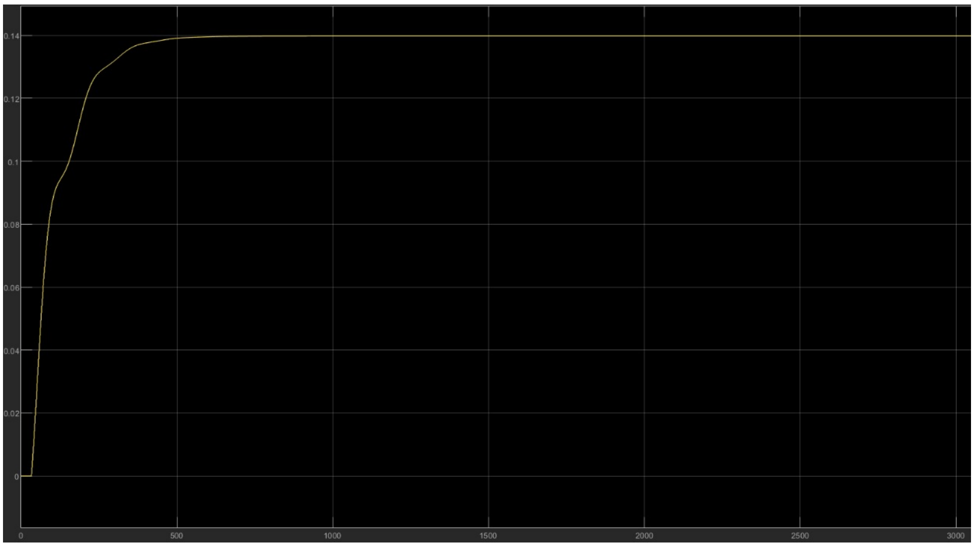

Figure I.22: Recycling flow (discrete-time)

After applying a step change at $t = 0$, the discrete-time responses exhibit a slightly larger overshoot and more pronounced damping than the continuous model. Tank 1's level peaks above 0.60 m before settling at 0.50 m, Tank 2 overshoots near 0.55 m to reach 0.47 m, and Tank 3 briefly spikes to 0.58 m before converging to 0.40 m. The inlet flow Q_{in} and recycling flow Q_{recycle} both show an initial surge followed by a damped approach to their steady-state values.

I.7.c Continuous-Time Relay Step Responses

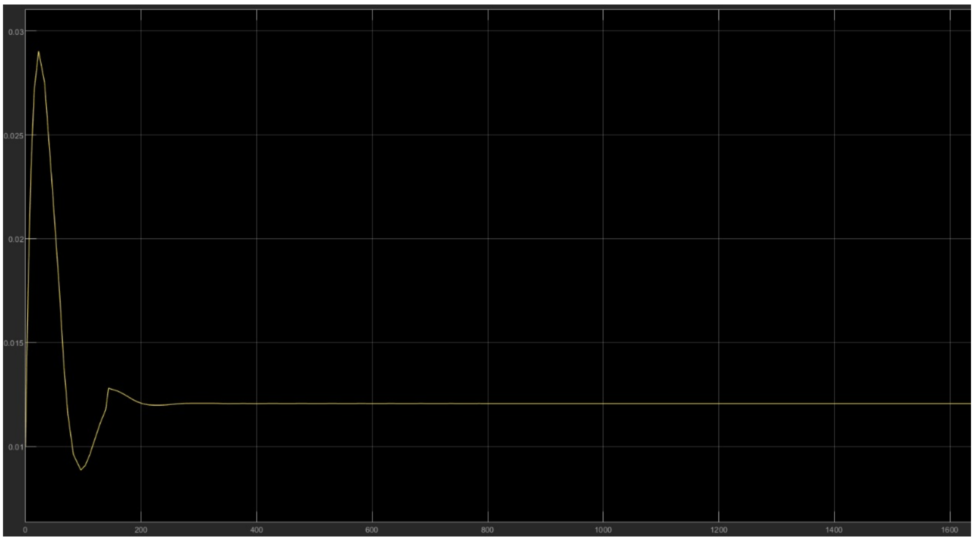

Figure I.23: Freshwater inflow (continuous-time relay)

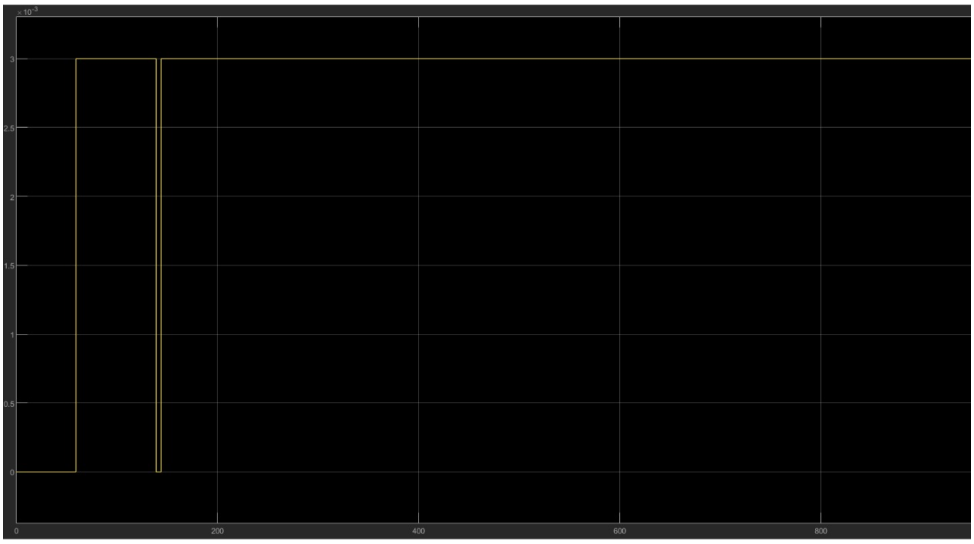

Figure I.24: Recycling flow (continuous-time relay)

Upon a level setpoint change, the continuous-time relay immediately toggles the pump: the inflow Q_{in} briefly drops to zero while the pump is off, then returns to its maximum when the level falls below the lower threshold. The recycling flow Q_{recycle} remains at its maximum whenever the relay is on, producing a square-wave response at the pump thresholds.

I.7.d Discrete-Time Relay Step Responses

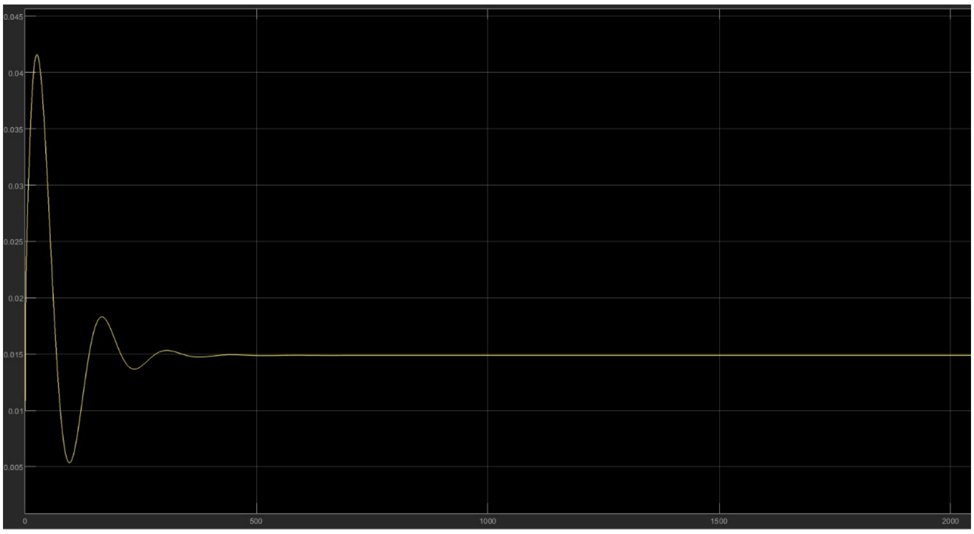

Figure I.25: Freshwater inflow (discrete-time relay)

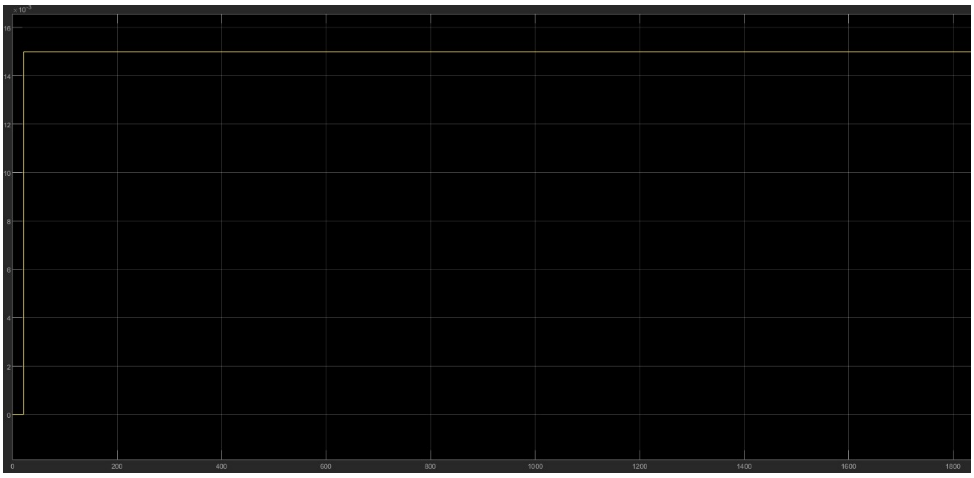

Figure I.26: Recycling flow (discrete-time relay)

In the discrete-time relay, the one-step delay causes a slight lag: the pump and inflow remain in their previous state for one sampling interval after the level crosses a threshold. As a result, both Q_{in} and Q_{recycle} exhibit delayed square-wave patterns compared to the continuous version, but still achieve the same on/off behaviour around the level limits.

I.8 Sustainability Analysis

I.8.a Freshwater Saved

Quantify the freshwater saved by calculating the recycling percentage:

$$\text{Recycling } \% = \frac{Q_{\text{recycle}}}{Q_{\text{in}} + Q_{\text{recycle}}}$$

- Continuous – PID : $Q_{\text{recycle}} = 2.959 \times 10^{-3}$ et $Q_{\text{in}} = 1.219 \times 10^{-2}$

$$\%\text{water\_recycling} = \frac{2.959 \times 10^{-3}}{1.219 \times 10^{-2} + 2.959 \times 10^{-3}} \times 100 \approx 19.5\%.$$

- Discrete – PID : $Q_{\text{recycle}} = 1.352 \times 10^{-2}$ et $Q_{\text{in}} = 1.231 \times 10^{-2}$

$$\%\text{water\_recycling} = \frac{1.352 \times 10^{-2}}{1.231 \times 10^{-2} + 1.352 \times 10^{-2}} \times 100 \approx 52.34\%.$$

- Continuous – Relay : $Q_{\text{recycle}} = 2.924 \times 10^{-3}$ et $Q_{\text{in}} = 1.223 \times 10^{-2}$

$$\%\text{water\_recycling} = \frac{2.924 \times 10^{-3}}{1.223 \times 10^{-2} + 2.924 \times 10^{-3}} \times 100 \approx 19.3\%.$$

- Discrete – Relay : $Q_{\text{recycle}} = 1.491 \times 10^{-2}$ et $Q_{\text{in}} = 1.512 \times 10^{-2}$

$$\%\text{water\_recycling} = \frac{1.491 \times 10^{-2}}{1.512 \times 10^{-2} + 1.491 \times 10^{-2}} \times 100 \approx 49.6\%.$$

I.8.b Energy Consumption - Formula

We first want to calculate the power of a pump. From the general relation

$$P_{pump} = \Delta p \cdot Q,$$

We want

$$P_{pump} = \rho\, g\, Q\, (h_1 - h_3).$$

With

$$\begin{aligned} h_1 &: \text{height of Tank 1} \\ h_3 &: \text{height of Tank 3} \\ \rho &: \text{water density} \\ g &: \text{gravitational acceleration} \\ Q &: \text{pump flow rate (specifically } Q_{\text{recycle}}) \end{aligned}$$

Proof:
For a fluid of constant density ρ, the gauge pressure at depth h is

$$p = \rho\, g\, h.$$

Let h_1 be the free-surface height at the pump outlet, and h_3 the height at the pump inlet. Then

$$\Delta p = p_{\text{out}} - p_{\text{in}} = \rho\, g\, h_1 - \rho\, g\, h_3 = \rho\, g\, (h_1 - h_3).$$

Substitute $\Delta p = \rho g(h_1 - h_3)$ into $P_{pump} = \Delta p\, Q$:

$$P_{pump} = \Big[\rho\, g\, (h_1 - h_3)\Big] \times Q = \rho\, g\, Q\, (h_1 - h_3)\,.$$

Get the Energy from the power
Right now, to get the energy used by the pump, we integrate the The energy consumed by the pump over one hour is given by the continuous integral

$$E_{\text{pump}} = \int_0^{3600} P_{\text{pump}}(t)\,\mathrm{d}t.$$

In the discrete model with time step Δt, this becomes

$$E_{\text{pump}} = \sum_{i=0}^{N-1} P_{\text{pump}}^{(i)}\,\Delta t, \quad N = \frac{3600}{\Delta t}.$$

To simplify the computation, we will use the average power because for any function $f : [a, b] \to \mathbb{R}$, let

$$m = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \quad \Longrightarrow \quad \int_a^b f(x)\,\mathrm{d}x = m\,(b-a).$$

I.8.c Energy Consumption

- Continuous - PID :

  $$\overline{Q_{recycle}\,(h_1 - h3)} = 2.959 \times 10^{-4}$$

  Therefore

  $$\overline{P} = \rho\, g\, \overline{Q_{recycle}\,(h_1 - h3)} = 1000 \times 9,81 \times 2.959 \times 10^{-4} = 2.903\,\mathrm{W}$$

  Finally,

  $$E_{pump} = \overline{P} \times 3600 = 10450.8\,\mathrm{J}$$

- Discrete - PID :

  $$\overline{Q_{recycle}\,(h_1 - h3)} = 1.325 \times 10^{-3}$$

  Therefore,

  $$\overline{P} = \rho\, g\, \overline{Q_{recycle}\,(h_1 - h3)} = 1000 \times 9,81 \times 1.325 \times 10^{-3} = 13\,\mathrm{W}$$

  Finally, with $\Delta t = 0.05$ and $N = \frac{3600}{0.05} = 72000$

  $$E_{pump} = \sum_{i=0}^{N-1} \overline{P} \times \Delta t = 13 \times 72000 \times 0.05 = 13 \times 3600 = 46800\,\mathrm{J}$$

- Continuous – Relay:

  $$\overline{Q_{\text{recycle}}\,(h_1 - h_3)} = 2.87 \times 10^{-4}$$

  Therefore,

  $$\overline{P} = \rho\, g\, \overline{Q_{\text{recycle}}\,(h_1 - h_3)} = 1000 \times 9.81 \times 2.87 \times 10^{-4} = 2.815 \text{ W}.$$

  Finally,

  $$E_{\text{pump}} = \overline{P} \times 3600 = 2.815 \times 3600 = 1.0136 \times 10^4 \text{ J}.$$

- Discrete – Relay:

  $$\overline{Q_{\text{recycle}}\,(h_1 - h_3)} = 5.816 \times 10^{-5}$$

  Therefore,

  $$\overline{P} = \rho\, g\, \overline{Q_{\text{recycle}}\,(h_1 - h_3)} = 1000 \times 9.81 \times 5.816 \times 10^{-5} \approx 0.571 \text{ W}.$$

  Finally, with $\Delta t = 0.05$ and $N = \frac{3600}{0.05} = 72000$,

  $$E_{\text{pump}} = \sum_{i=0}^{N-1} \overline{P}\,\Delta t = 0.571 \times 72000 \times 0.05 = 0.571 \times 3600 \approx 2055 \text{ J}.$$

I.8.d Summary

Overall, the system achieves a trade-off between recycling rate and energy use. The continuous-time PID recovers about 19.5 % of inflow at a pump energy cost of 10.45 kJ/h, while discrete-time PID pushes recycling above 50 % but consumes 46.8 kJ/h. Relay control yields lower recycling ($\approx$ 19 % continuous, 49.6 % discrete) and the lowest energy use ($\approx$ 10.1 kJ/h continuous, 2.06 kJ/h discrete). These figures allow choosing the best balance for a given operation.

I.9 Conclusion

We began by formulating the nonlinear mass-balance equations for three interconnected tanks, capturing the $\sqrt{\Delta h}$ coupling that defines our recycling loop. After establishing system parameters and discretizing via explicit Euler, we validated the discrete model against an ODE solver, confirming accurate replication of continuous dynamics.

In the control design phase, we implemented freshwater inflow regulation with a PID loop on Tank 1 and compared both PID and relay strategies on Tank 3. Continuous-time PID yielded smooth level regulation with moderate overshoot and a recycling rate near 20 %, at a modest energy cost. Discrete-time PID greatly increased reuse (50 %) but at a substantially higher pump energy—highlighting the trade-off between water savings and power draw. Relay control, both continuous and sampled, delivered simple on/off operation with lower energy use but more oscillatory levels and only moderate recycling gains.

Simulation results confirmed these behaviors: step responses for all three tanks showed how tuning and sampling affect overshoot, settling time, and steady-state accuracy. The sustainability analysis quantified fresh-water savings and pump energy, providing clear metrics for decision-making.

In summary, our study demonstrates a flexible framework: by swapping control schemes within each tank's Simulink mask, operators can target high reuse, low energy, or fast settling as needed. The nonlinear, coupled dynamics were handled effectively by both PID and relay, and the discrete implementation faithfully reproduced continuous behavior while allowing straightforward digital deployment. This comprehensive approach—from modeling through sustainability analysis—provides a robust basis for real-world industrial water recycling.